

Below is a **detailed lab manual** to set up a **PersistentVolume (PV)** and **PersistentVolumeClaim (PVC)** on a **Minikube** cluster and run an **NGINX Deployment** that uses that storage.

Lab: PV + PVC on Minikube with NGINX

Goal

1. Create a **hostPath-based PV** (works well for Minikube labs).
2. Create a **PVC** that binds to that PV.
3. Deploy **NGINX** and mount the PVC at `/usr/share/nginx/html`.
4. Write a file into the mounted storage and verify it survives pod reschedule.

Prerequisites

- Minikube installed
 - kubectl installed
 - Minikube running
-

0) Start Minikube and verify cluster

```
minikube start
```

```
kubectl cluster-info
```

```
kubectl get nodes
```

(Optional) Ensure default storage class exists:

```
kubectl get sc
```

1) Create a dedicated namespace (recommended)

```
kubectl create ns storage-lab
```

```
kubectl config set-context --current --namespace=storage-lab
```

```
kubectl get ns
```

2) Create a directory inside Minikube node for PV storage

Minikube runs Kubernetes nodes inside a VM/container. We'll create the path **inside the Minikube node**.

```
minikube ssh "sudo mkdir -p /mnt/data/nginx-pv && sudo chmod 777 /mnt/data/nginx-pv"
```

```
minikube ssh "ls -ld /mnt/data/nginx-pv"
```

✅ This directory will be the PV backing storage.

3) Create the PersistentVolume (PV)

Create a file: pv.yaml

```
apiVersion: v1
```

```
kind: PersistentVolume
```

```
metadata:
```

```
  name: nginx-pv
```

```
spec:
```

```
  capacity:
```

```
    storage: 1Gi
```

```
  accessModes:
```

```
    - ReadWriteOnce
```

```
  persistentVolumeReclaimPolicy: Retain
```

```
  storageClassName: manual
```

```
  hostPath:
```

```
    path: /mnt/data/nginx-pv
```

Apply it:

```
kubectl apply -f pv.yaml
```

```
kubectl get pv
```

```
kubectl describe pv nginx-pv
```

Expected:

- PV status initially: **Available**
-

4) Create the PersistentVolumeClaim (PVC)

Create a file: pvc.yaml

apiVersion: v1

kind: PersistentVolumeClaim

metadata:

name: nginx-pvc

spec:

accessModes:

- ReadWriteOnce

resources:

requests:

storage: 1Gi

storageClassName: manual

Apply it:

kubectl apply -f pvc.yaml

kubectl get pvc

kubectl describe pvc nginx-pvc

Expected:

- PVC status should become: **Bound**
- PV status should become: **Bound**

Verify binding:

kubectl get pv,pvc

5) Deploy NGINX using the PVC

Create a file: nginx-deploy.yaml

apiVersion: apps/v1

kind: Deployment

metadata:

name: nginx-deploy

spec:

replicas: 1

selector:

matchLabels:

app: nginx-pv

template:

metadata:

labels:

app: nginx-pv

spec:

containers:

- name: nginx

image: nginx:stable

ports:

- containerPort: 80

volumeMounts:

- name: nginx-storage

mountPath: /usr/share/nginx/html

volumes:

- name: nginx-storage

persistentVolumeClaim:

claimName: nginx-pvc

Apply:

kubectl apply -f nginx-deploy.yaml

kubectl get deploy,pods

Wait until pod is Running:

kubectl get pods -w

6) Expose NGINX with a Service

Create a file: nginx-svc.yaml

apiVersion: v1

kind: Service

metadata:

name: nginx-svc

spec:

type: NodePort

selector:

app: nginx-pv

ports:

- port: 80

targetPort: 80

Apply:

kubectl apply -f nginx-svc.yaml

kubectl get svc nginx-svc

Open it:

minikube service nginx-svc -n storage-lab

7) Validate the PVC mount by writing content

Get pod name:

```
kubectl get pods
```

Exec into the pod and create a test page:

```
POD=$(kubectl get pod -l app=nginx-pv -o jsonpath='{.items[0].metadata.name}')
```

```
kubectl exec -it $POD -- sh -c 'echo "Hello from PV/PVC $(date)" >
/usr/share/nginx/html/index.html && cat /usr/share/nginx/html/index.html'
```

Now test from your browser (or curl):

```
minikube service nginx-svc -n storage-lab --url
```

Copy the URL and:

```
curl $(minikube service nginx-svc -n storage-lab --url)
```

You should see your “Hello from PV/PVC ...” text.

8) Persistence test (delete pod and verify data stays)

Delete the pod (deployment will recreate it):

```
kubectl delete pod -l app=nginx-pv
```

```
kubectl get pods -w
```

After the new pod is Running:

```
NEWPOD=$(kubectl get pod -l app=nginx-pv -o jsonpath='{.items[0].metadata.name}')
```

```
kubectl exec -it $NEWPOD -- cat /usr/share/nginx/html/index.html
```

✅ If the text is still there, persistence is working.

9) Verify data exists on Minikube node

Check directly on Minikube node path:

```
minikube ssh "ls -l /mnt/data/nginx-pv && cat /mnt/data/nginx-pv/index.html"
```

10) Cleanup (optional)

If you want to remove everything:

```
kubectl delete -f nginx-svc.yaml
```

```
kubectl delete -f nginx-deploy.yaml
```

```
kubectl delete -f pvc.yaml
```

```
kubectl delete -f pv.yaml
```

```
kubectl delete ns storage-lab
```

If you used Retain, PV data stays on disk. Remove it:

```
minikube ssh "sudo rm -rf /mnt/data/nginx-pv"
```

Troubleshooting Guide

PVC stuck in Pending

Check:

```
kubectl describe pvc nginx-pvc
```

```
kubectl get pv
```

Common causes:

- storageClassName mismatch between PV and PVC (both must be manual here).
 - PV capacity smaller than PVC request.
 - accessModes mismatch.
-

Pod stuck in ContainerCreating

Check events:

```
kubectl describe pod <pod-name>
```

Common cause:

- hostPath directory does not exist in minikube node (/mnt/data/nginx-pv). Recreate with minikube ssh.

Permission denied in NGINX html directory

Fix permissions on the node:

```
minikube ssh "sudo chmod -R 777 /mnt/data/nginx-pv"
```